

Statistical & Numerical Methods HW4

Daniel Haim Breger, 316136944

Technion

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1 Estimators

We know

$$\mu = 5, \quad \sigma^2 = 0.05 \quad (1)$$

1.1 length estimator

The estimator is just

$$\hat{\mu} = \bar{x} = 5.512 \quad (2)$$

1.2 variance estimator

using the formula for an estimator with an unknown μ

$$\sigma^2 = \frac{1}{N-1} \sum_i (x_i - \bar{x})^2 = 0.052 \quad (3)$$

1.3 CL95

We'll assume the distribution of lengths is a normal distribution, so the standard deviation is

$$\sigma' = \frac{\sigma}{\sqrt{N}} = \frac{\sqrt{0.052}}{\sqrt{10}} = 0.072 \quad (4)$$

so

$$\mu_+ = 1.96\sigma' + \mu = 5.653, \quad \mu_- = 5.371 \quad (5)$$

1.4 Brown screws

directly the expected p is

$$p = \frac{b}{N} = 0.6 \quad (6)$$

and for the variance, marking the number of brown screws as b ,

$$\sigma_p^2 = \left(\frac{\partial p}{\partial b} \right)^2 \sigma_b^2 = \frac{\sigma_b^2}{N^2} \quad (7)$$

the number of brown screws pulled is binomial so

$$\sigma_p^2 = \frac{Np(1-p)}{N^2} = \frac{p(1-p)}{N} = \frac{0.6 \cdot 0.4}{100} = 0.0024 \quad (8)$$

so

$$\sigma_p = 0.049 \quad (9)$$

so the CL95 range is

$$\mu_- = p - 1.96\sigma_p = 0.504, \quad \mu_+ = 0.696 \quad (10)$$

2 Radioactive Decay

2.1 5 minutes

The decay rate is known so the different minutes are independent meaning

$$N_5 = 5N_1 \quad (11)$$

but since N_1 is distribution poisson,

$$N_1 = \lambda \pm \sqrt{\lambda} \quad (12)$$

so

$$N_5 = 5\lambda \pm \sqrt{5\lambda} = 500 \pm 10\sqrt{5} \quad (13)$$

2.2 Unknown material

Since we assume the decay is poisson this tells us that

$$\lambda = 100 \quad (14)$$

which implies

$$\sigma = \sqrt{\lambda} = 10 \quad (15)$$

which is a variance whose source is the measurement. We add this to the variance from the last part which is inherent to the decay and get

$$N = 500 \pm 10\sqrt{5} \pm 50 \quad (16)$$

2.3 composite material

Since we measured the number of decays of both materials together and they are independent we measured

$$N_i \sim Pois(\lambda_1 + \lambda_2) \quad (17)$$

so this is not enough to measure each separately. Any attempt to create an equation that separates them will fail, and we'll always have dependence on $\lambda_1 + \lambda_2$.

3 Basketball

3.1 random variables

Who's Shimshon? I'm assuming this is supposed to be Gal :)

We can define S_1, S_2, S_3 to be the score achieved through penalties, inside, and outside throws. This means the total score he earned is

$$S = S_1 + S_2 + S_3 \quad (18)$$

but the number of successful throws is distributed binomial so

$$S_1 \sim \text{Bin}(N_1, q_1), \quad S_2 \sim \text{Bin}(N_2, q_2), \quad S_3 \sim \text{Bin}(N_3, q_3) \quad (19)$$

noting the large N s are different from small n s. N represents an unknown number of attempts for the distribution, n is the number actually attempted.

3.2 Average score

Who's Shimshon? I'm assuming this is supposed to be Gal :)

$$E[S] = E[S_1 + 2S_2 + 3S_3] = E[S_1] + 2E[S_2] + 3E[S_3] \quad (20)$$

but

$$E[S_1] = n_1 q_1 = 6 \cdot 0.75 = 4.5, \quad E[S_2] = 6, \quad E[S_3] = 4 \quad (21)$$

so

$$E[S] = 28.5 \quad (22)$$

and the variance is, since each set is independent of the others

$$V[S] = V[S_1 + 2S_2 + 3S_3] = V[S_1] + 4V[S_2] + 9V[S_3] = n_1 q_1 (1 - q_1) + 4n_2 q_2 (1 - q_2) + 9n_3 q_3 (1 - q_3) \quad (23)$$

$$= 19.08 \implies \sigma = 4.37 \quad (24)$$

3.3 Most probable throws

The assumption H_0 implies

$$0.75S_1 + 2 \cdot 0.75S_2 + 3 \cdot 0.66S_3 = 24 \quad (25)$$

together with

$$0.25S_1 + 0.25S_2 + 0.34S_3 = 5 \quad (26)$$

and

$$S_1 + S_2 + S_3 = 19 \quad (27)$$

which is three equations for three variables so

$$\begin{pmatrix} 1/2 & 3/2 & 2 \\ 1/4 & 1/4 & 1/3 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} S_1 \\ S_2 \\ S_3 \end{pmatrix} = \begin{pmatrix} 24 \\ 5 \\ 19 \end{pmatrix} \quad (28)$$

throwing into wolfram alpha we get

$$S_1 = 8, \quad S_2 = 8, \quad S_3 = 3 \quad (29)$$

3.4 Extension

Who's Shimshon? I'm assuming this is supposed to be Gal :)

after the extension the numbers are

$$S_1 = 10, \quad S_2 = 8, \quad S_3 = 6 \quad (30)$$

so the p-value is

$$P(S_1 < 8 \cdot 0.75, S_2 < 8 \cdot 0.75, S_3 < 3 \cdot 0.66 | H_0) \quad (31)$$

$$= P(S_1 < 6, S_2 < 6, S_3 < 2 | H_0) \quad (32)$$

$$= P(S_1 < 6)P(S_2 < 6)P(S_3 < 2) \quad (33)$$

which can be directly computed using binomial distribution probability distributions.

4 Signal and Noise

The error on the parameters means

$$b = 8 \pm 0.56, \quad \phi = 0.5 \pm 0.035, \quad \omega = \frac{1}{18} \pm 0.004 \quad (34)$$

4.1 signal and noise

we can find the amount of noise by integrating over the interesting regions

$$\int dx B = b \left(x - \frac{\cos(\omega x + \phi)}{\omega} \right) + C \quad (35)$$

so

$$B_1 = \int_{-14}^{-5} B dx = b \left(9 - \frac{\cos(-5\omega + \phi)}{\omega} + \frac{\cos(-14\omega + \phi)}{\omega} \right) = 70.02 \quad (36)$$

$$B_2 = \int_5^{14} B dx = b \left(9 + \frac{\cos(5\omega + \phi)}{\omega} - \frac{\cos(14\omega + \phi)}{\omega} \right) = 133 \quad (37)$$

and the errors

$$\sigma_{B_1}^2 = \left(\frac{\partial B_1}{\partial b} \right)^2 \sigma_b^2 + \left(\frac{\partial B_1}{\partial \omega} \right)^2 \sigma_\omega^2 + \left(\frac{\partial B_1}{\partial \phi} \right)^2 \sigma_\phi^2 \quad (38)$$

$$= (8.72\sigma_b)^2 + (-676\sigma_\omega)^2 + (71\sigma_\phi)^2 = 37.55 \quad (39)$$

and

$$\sigma_{B_1}^2 = \left(\frac{\partial B_1}{\partial b} \right)^2 \sigma_b^2 + \left(\frac{\partial B_1}{\partial \omega} \right)^2 \sigma_\omega^2 + \left(\frac{\partial B_1}{\partial \phi} \right)^2 \sigma_\phi^2 \quad (40)$$

$$= (16.63\sigma_b)^2 + (327\sigma_\omega)^2 + (37\sigma_\phi)^2 = 90.05 \quad (41)$$

so

$$B_1 = 70.02 \pm 6.13, \quad B_2 = 133 \pm 9.49 \quad (42)$$

so the signal is

$$S_1 = N_1 - B_1 = 84 - 70.02 = 13.98 \pm 6.13 \pm \sqrt{13.98}, \quad S_2 = 147 - 133 = 14 \pm 9.49 \pm \sqrt{14} \quad (43)$$

and

$$seg_1 = \frac{S_1}{\sqrt{B_1}} = 1.67, \quad seg_2 = \frac{S_2}{\sqrt{B_2}} = 1.21 \quad (44)$$